

The Netflix Buffering Problem

Slide 1: The Netflix Buffering Problem (Hook + Reality)

 Friday Night, Peak Hour...

It's Friday night, peak internet hours.

You sit down to watch the most intense episode of your favorite Netflix series.

The story builds. The music rises.

And then...

BUFFERING... 25% 

You wait.

The video resumes.

Thirty seconds later — BUFFERING AGAIN.

 **Why This Moment Matters:**

This isn't just a small inconvenience.

For users, buffering:

- Breaks emotional engagement
- Ruins key moments of the story
- Creates frustration and impatience

For Netflix, buffering means:

- Users leaving the platform
- Lower satisfaction
- Reduced watch time and retention

Visual Insight

[PLACEHOLDER: Insert visual diagram showing viewer drop after buffering events]

 A sharp drop in viewers after repeated buffering events proves that playback smoothness is more important than video quality.

Slide 2: Why Buffering Happens (Core Problem)

Why can't Netflix simply fix buffering with one smart algorithm?

The answer lies in user diversity. Netflix users vary greatly in terms of:

- Internet speed
- Mobile data vs Wi-Fi usage
- Battery level
- Peak vs off-peak hours
- Device capability
- Network stability

Despite this diversity, many systems rely on a single automatic quality-selection algorithm. One fixed approach cannot satisfy all users, leading to buffering, frustration, and viewer abandonment.

This design challenge sets the stage for introducing the Strategy Design Pattern.